

Agricultural Analysis and Prediction Using Satellite Imagery

Abstract—Satellite imagery has brought about a transformative shift in the monitoring and management of agricultural practices through advanced analysis and predictive capabilities. Satellite imagery provides a comprehensive and timely assessment of various agricultural parameters such as crop health, vegetation indices, soil moisture content, and land cover classification. Satellites cover large areas consistently and frequently, providing data continuity for monitoring changes over time, identifying trends, and predicting crop yields or potential failures. This research specifically focuses on the classification of different types of crop lands from satellite images. The study encompasses 10 distinct types of crop lands, utilizing satellite imagery spanning from 2017 to 2020, thereby representing varying crop field conditions over time. To achieve higher classification accuracy than existing models, Convolutional Neural Networks (CNNs) were employed as the primary machine learning technique. The CNN architecture comprises multiple convolutional layers and corresponding pooling layers, forming a comprehensive pipeline for data preprocessing and classification.

Index Terms—Convolutional Neural Networks, Convolutional Layer, Satellites, Agricultural patterns

I. INTRODUCTION

In recent years, advancements in satellite technology have revolutionized various fields, and agriculture is no exception. Satellite imagery has emerged as a powerful tool for analyzing and predicting agricultural patterns, providing valuable insights to farmers, researchers, and policymakers. This thesis aims to explore the potential of satellite imagery in agricultural analysis and prediction, delving into its applications, benefits, and challenges. By harnessing the vast amount of data captured by satellites, we can unlock a deeper understanding of crop health, soil moisture, yield estimation, and pest management. The integration of satellite imagery with machine learning algorithms offers unprecedented opportunities to enhance decision-making processes and optimize agricultural practices. Our research, aims to classify crop land of different types from satellite images. In this research we particularly worked with 10 different types of crop land and the satellite imagery were taken from a a time period of 2017 to 2020. It reflects different crop field i their different from. For our research, we used Convolutional Neural Network and classified the crop lands to achieve a better accuracy then current models. The architecture includes multiple Convolutional layers and corresponding pooling layers which eventually made up the complete Convolutional pipeline from data preprocessing to classification. Through this research, we strive to contribute to the advancement of precision agriculture, ultimately leading to increased productivity, sustainability, and food security.

II. LITERATURE REVIEW

The utilization of high-resolution satellite images is a focal point of study in this field. The utilization of satellite technology in agricultural phenotyping demonstrates significant implications. The significance of symbolism in agricultural creation, the process of yield phenotype, and its implications. This study by Zhang. et al suggests plant characteristics that may be evaluated using high-precision measurement techniques. Satellite imagery refers to the visual representation of the Earth's surface or other celestial bodies captured by satellites orbiting the planet. The study provides a comprehensive overview of the advantages (an illustration of expeditious or automated data extracted from a larger source and diverse field locations) and challenges, such as the primary obstacle (cloud hindrance) faced by satellite-based phenotyping in agricultural research: the presence of clouds [1]. The production of programs and the examination of future perspectives and possibilities The concept of high-goal satellite symbolism may be understood as a phenomenon characterized by its high-throughput nature. A writing tool used for inputting text or characters onto a surface. Also this paper examines the use of satellite imagery, particularly WorldView-3 as a high-resolution source, while acknowledging the limitations posed by cloud cover and suggesting the potential of synthetic-aperture radar (SAR) technology as a cloud-independent alternative [2].

In this paper by Sharifi. et al an agricultural field extraction by the use deep learning algorithm along with Satellite Imagery used ResUNet-a to extract field borders [3]. Also, Sentinel-2 is used with geometrical accuracy and it completed a progression of examinations that show that overfitting has been limited without re-adjustment to be executed in different circumstances. In this paper, the ResUNet-a D6 and D7 model was trained by Adam as the optimizer and it accomplishes quicker assembly than different other options [4].

Optical satellite imaging within the visible to near-infrared spectrum is frequently more conducive to the interpretation and classification of water bodies and flooded regions in comparison to radar pictures. From the paper by Schumann et al. the utilization of aerial photography characterized by a significant level of spatial resolution enables the derivation of flood extent through the process of digitizing the boundaries seen at the interface between land and water. The Normalized Difference Water Index (NDWI) is a widely used technique in the field of image processing, which aims to increase the visibility of open water surfaces while simultaneously suppressing the presence of soil and vegetation characteristics. Optical

satellite imagery has shown to be a significant tool for the mapping of floods for a period exceeding four decades, owing to the existence of a substantial historical repository. The sensors known as Moderate Resolution Imaging Spectroradiometer (MODIS) developed by NASA, another technology VIIRS (Visible Infrared Imaging Radiometer Suite) developed by NOAA (National Oceanic and Atmospheric Administration) offer comprehensive worldwide coverage and play a crucial role in the surveillance of flood occurrences. Nonetheless, the efficacy of flood detection with optical sensors is hindered by factors such as prolonged cloud cover, the presence of vegetation, and heavily populated urban areas. The effectiveness of optical imaging is impeded by the presence of cloud cover, rendering it incapable of accurately mapping flood occurrences in metropolitan regions or beneath dense plant canopies [5].

This research work by Roznik et al. makes a theoretical addition by examining the impact of employing higher-resolution satellite images and an extensive agricultural mask on the precision of calculating crop yields at the county level. Satellite imaging is a valuable and pertinent source of extensive data that exhibits potential for many applications within the agricultural domain. When employing satellite images for the purpose of calculating agricultural productivity, practitioners must make judgments on the chosen picture resolution. The act of choosing higher-resolution photographs requires a larger allocation of computational resources as compared to lower-resolution alternatives. Although using lower-resolution pictures might lead to computational cost reductions, it may also lead to a decrease in the accuracy of crop yield models. The economic feasibility of satellite image processing has been greatly improved due to the progress made in cloud computing and the availability of open-access computing platforms, such as Google Earth Engine. The primary aim of this research is to evaluate the potential enhancement in crop yield accuracy by the use of higher resolution normalized difference vegetation index (NDVI) data with a cropland mask [6]. The research conducted involved the compilation of Normalized Difference Vegetation Index (NDVI) measurements. These measures are widely used as a reliable indication of crop greenness. The study focused on four specific crop kinds, namely maize, soybeans, spring wheat, and winter wheat. The data collection spanned throughout 48 states within the United States. The data was gathered during a duration of 11 years. The regression models for crop production shown an improvement in the value of R^2 because of the resolution of the NDVI data the satellite had increased. The predictions indicate that the utilization of satellite NDVI data with higher resolution leads to more accurate predictions of crop yield as compared to the NDVI data of the satellite with lower resolution. The study gathered Normalized Difference Vegetation Index (NDVI) measurements, which serve as an indicator of crop greenness, for four distinct crop types (corn, soybeans, spring and winter wheat) throughout 48 states in the United States during a span of 11 years. The R^2 value, which represents the accuracy of yield of each crop in regression model, showed improvement with increasing resolution of the satellite NDVI data. The

findings suggest that employing satellite NDVI data with better resolution yields more precise estimations of crop production in comparison to satellite NDVI data with lower resolution. This research is widely regarded as the most extensive investigation conducted thus far on the utilization of NDVI for the calculation of agricultural production. It encompasses data from 48 states in the United States, four distinct crops, and three varying degrees of resolution, spanning a duration of 11 years [7].

The research by Haque et al. provides a theoretical contribution by investigating the effects of employing higher resolution imagery of satellite and a comprehensive agricultural mask on the accuracy of estimating yield of different crops. Satellite imagery is a significant and useful source of comprehensive data that has the potential for a variety of agricultural applications. When utilizing satellite imagery for the estimation of agricultural production, practitioners must make decisions regarding the image resolution they will employ. The selection of photographs with a higher resolution requires a larger allocation of computer resources than the selection of photographs with a lower resolution. Choosing lower-resolution images may reduce computational costs, but crop yield models could become less accurate as a result. Advances in cloud computing and the availability of open-access computing platforms have increased the economic viability of satellite image processing. This study aims to determine the degree to which crop yield estimation precision can be enhanced by incorporating higher resolution normalized difference vegetation index (NDVI) data along with croplands overlay. The study compiled Normalized Difference Vegetation Index (NDVI) measurements, which serve as an indicator of crop greenness, for four distinct crop types (soybeans, winter wheat ,corn, spring wheat) in 48 U.S. states. This data was accumulated over the course of eleven years. The R^2 value of each crop yield regression model increased as satellite data resolution increased. The findings suggest that using NDVI data with a better resolution generates more precise estimates of crop production than using satellite NDVI data with a lower resolution. The research collected NDVI data, which measures the greenness of crops, for four commodities (soybeans, winter wheat ,corn, spring wheat) in 48 U.S. states over an 11-year period. As the NDVI satellite data resolution increased, the accuracy (R^2) of yield of crop in regression model had improved. The results indicate that higher-resolution satellite NDVI data results in better accuracy for crop yield estimation than any NDVI data from lower resolution satellite . Covering four crops differnt crops from forty eight U.S. states, and three distinct levels of resolution over an 11-year period, this research is considered to be the very comprehensive investigation of date into the application of NDVI for crop yield estimation [8].

Exploring the utilization of machine learning and deep learning strategies for categorizing sequences of satellite images, specifically focusing on mapping crops. Traditional techniques for analyzing such image sequences depended on manually designed characteristics and domain expertise, which necessitated time-intensive feature engineering and yielded lim-

ited performance. Conversely, deep learning-based approaches have displayed encouraging outcomes by autonomously learning intricate attributes from unprocessed data.

The typical deep learning methods for crop mapping involve classifying individual pixels through 1D Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) [9], [10]. However, these methodologies often overlook the spatial correlation between other pixels around it and chronological correlation within image sequences. To address these constraints, a 3D Fully Convolutional Neural Network (FCN) is adopted, allowing the model to grasp both the spatial and temporal connections within image sequences.

The article by Mohammadi et al. also introduces the concept of supervising intermediary layers within deep neural networks. While most techniques solely employ output supervision for network training, various studies have demonstrated that overseeing intermediate layers can enhance segmentation outcomes. The paper presents two strategies for intermediary supervision: Cross-entropy loss Middle Supervision (CE-MidS) and Supervised Contrastive loss Middle Supervision (SupCon-MidS). These approaches strive to generate more distinct attributes for each type of crop, resulting in more precise predictions [11].

III. DATASET

The objective of our study was to employ the dataset presented in the scholarly article titled "The Canadian Cropland Dataset: A New Land Cover Dataset for Multitemporal Deep Learning Classification in Agriculture." The dataset comprises the ten crops that are most often farmed in Canada.

Crop Type	Image Count
Barley	5382
Canola	11366
Corn	12878
Mixed-Wood	4981
Oats	4807
Orchards	6594
Pasture and Forages	8797
Potatoes	5294
Soybean	10208
Spring Wheat	8229

Fig. 1. Image per class

As illustrated in Fig. 1 the image amounts associated with each crop type are as follows: mixed-wood (4,981), oats (4,807), barley (5,382), potatoes (5,294), orchards (6,594), spring wheat (8,229), pasture and forages (8,797), canola (11,366), soybean (10,208), and corn (12,878). Among the 6,633 defined geographical sites, a comprehensive collection of 93,175 photos was successfully obtained. The observed count is lower than the expected count of 132,660 temporal samples (calculated as 5 times 6,633 times 4) over the span of four years.

IV. MODEL AND ARCHITECTURE

The primary objective of the study is to design an image classification pipeline using deep learning techniques for

classifying cropland types from satellite imagery. The dataset consists of satellite image patches captured by Sentinel-2A for the purpose of cropland classification. The dataset is divided into three subsets: training, validation, and test sets. The research employs Convolutional Neural Networks (CNNs) due to their capability to capture spatial patterns within images effectively. A CNN consists of convolutional and pooling layers, which learn filters for feature extraction, and reduce dimensionality, respectively.

This represents the initial layer of the network. The architecture comprises a set of 32 convolutional filters, wherein each filter is responsible for scanning the input picture to detect and extract certain characteristics. The activation function known as Rectified Linear Unit (ReLU) is utilized to incorporate non-linear characteristics into the model, enabling it to acquire knowledge of intricate relationships present within the data [12]. Following the initial convolutional layer, a subsequent layer of max-pooling is implemented. The down-sampling process in this layer involves selecting the greatest value inside each 2x2 region of the feature maps generated by the preceding layer. This reduces the spatial dimensions, making the subsequent layers more computationally efficient and less prone to overfitting. Dropout is a regularization technique that helps prevent overfitting. During training, this layer randomly "drops out" (sets to zero) a fraction (in this case, 25%) of the neurons in the previous layer. This encourages the network to rely on multiple pathways for learning, making it more robust.

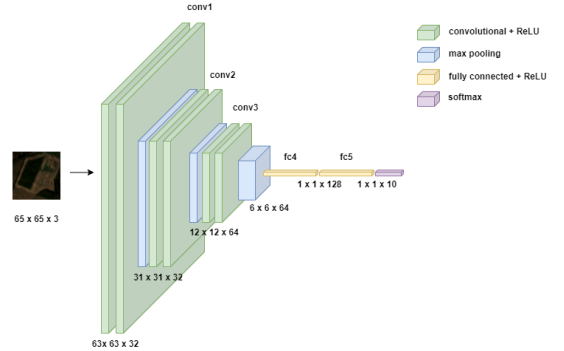


Fig. 2. Methodology

Like the initial convolutional layer, this layer is equipped with 64 filters and utilizes the rectified linear unit (ReLU) activation function. The objective is to acquire a deeper understanding of intricate and advanced characteristics derived from the dataset. A subsequent max-pooling layer is implemented, with the identical pool size as the previous layer, in order to further reduce the spatial dimensions of the feature maps. Another layer of dropout is added, with the dropout rate being identical to that of the initial layer. This regularization technique serves the purpose of preventing the model from excessively conforming to the training data.

An additional convolutional layer is implemented, consisting of 64 filters and utilizing the Rectified Linear Unit (ReLU)

activation function. This enables the model to get a greater understanding of intricate aspects. A subsequent max-pooling layer is employed, with an identical pool size, to further reduce the resolution of the feature maps. The subsequent layer is responsible for transforming the two-dimensional feature maps obtained from the preceding layer into a one-dimensional vector. The aforementioned transformation is important in order to establish a connection between the convolutional layers and the fully linked layers. A dropout layer featuring an increased dropout rate of 50 percent.

This layer offers a more robust kind of regularization.

The current layer under consideration is a fully linked layer consisting of 128 units. The ReLU activation function is applied to this layer. The algorithm extracts prominent characteristics from the compressed dataset.

The final layer consists of a fully linked layer with 10 units, which aligns with the total number of classes involved in the classification job, specifically pertaining to different types of farmland. The softmax activation function is utilized to transform the output of a neural network into a probability distribution across several classes, hence facilitating the model's capacity to provide predictions.

V. RESULT ANALYSIS AND FINDINGS

As illustrated in Fig. 3 the confusion matrix shown has a size of 10 by 10. Each cell representing each class accordingly. The diagonal cells represent the true positives which are the instances where the model correctly predicted the specific character. The confusion matrix of 10 by 10 presents the count of true positives, false positives, true negatives, and false negatives for every category in the set of assessments. This feature can prove to be beneficial in identifying the specific classes that the model encounters difficulty in accurately classifying.

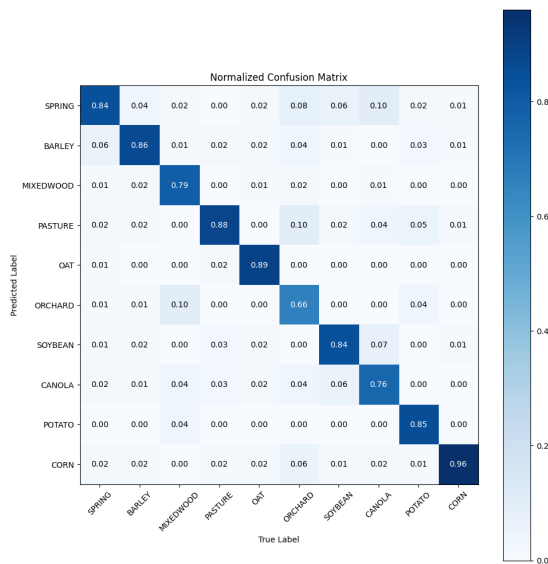


Fig. 3. Confusion Matrix

The classification report shown in Fig. 4 shows that the model worked pretty good for Spring, Barley, Pasture, Canola, and Corn. But in Mix wood, OAT, Orchard, and Potato the precision is a little off. Which means the model did some false positive. One reason for that could be complexity of the model wasn't high enough to accurately get the features that could identify these class with less false negative.

	Class	Precision	Recall	F1-Score	Support
0	SPRING	0.890000	0.710000	0.790000	1245
1	BARLEY	0.810000	0.810000	0.810000	656
2	MIXEDWOOD	0.620000	0.920000	0.740000	249
3	PASTURE	0.860000	0.770000	0.810000	733
4	OAT	0.670000	0.970000	0.790000	249
5	ORCHARD	0.390000	0.800000	0.520000	352
6	SOYBEAN	0.680000	0.840000	0.750000	460
7	CANOLA	0.790000	0.780000	0.780000	1116
8	POTATO	0.580000	0.960000	0.720000	188
9	CORN	0.890000	0.840000	0.910000	1950

Fig. 4. Classification Report

Using the LIME library we were able to understand how did the Convolutional Neural Network arrives at its predictions. This helps us to understand the in depth implementation of the model and how it uses its convolutional layers to filters the features. As illustrated in Fig. 5 the code overlays a mask on the image to highlight the features that contributes most to the prediction, providing visual insights into the decision-making process of the model.

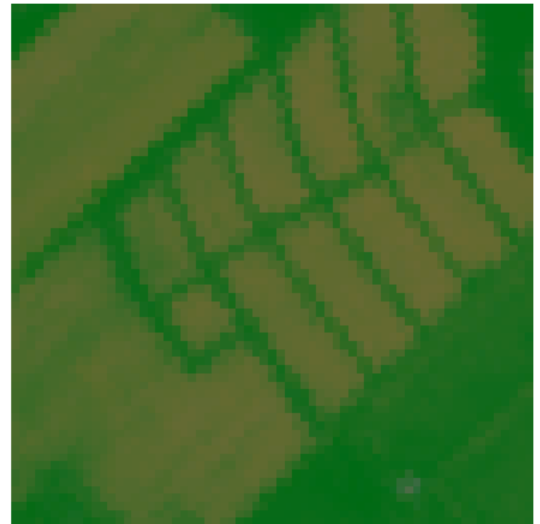


Fig. 5. LIME implementation

In summary, the current research presents promising results, characterised by a significant level of accuracy and a balanced F1 measure. Additional examination is required in order to

fully understand the variables that influence the efficacy of the model and to make comparisons with other state-of-art models. Moreover, it would be beneficial to explore the possible limitations of the model and identify areas for improvement.

VI. LIMITATIONS AND SCOPE

Satellite imagery can be affected by cloud cover, atmospheric interference, or variations in lighting conditions. These factors can impact the quality and availability of data, leading to incomplete or noisy observations.

Accurate ground truth data for validating and training models can be challenging to obtain, especially for large-scale agricultural prediction tasks

Training machine learning models typically requires labeled or annotated data for supervised learning. Labeling large-scale satellite imagery data can be time-consuming and costly.

Developing real-time monitoring systems that combine satellite imagery with near-real-time data streams can enable early detection of crop diseases, pest outbreaks, or adverse weather conditions Integrating satellite imagery data with other relevant data sources, such as weather data, soil data, or historical records, can enhance the predictive capabilities Integrating with advance IOT would conclude in better agricultural progress and opens up more research field.

We also implemented other models such as lstm, one type of recurrent neural network that implements different gates takes other units weight to work as a sequence. This model achieve an overall accuracy of 67 percentage. In transfer learning we used RESNET50, VGG16 and VGG19 which performed really good in traning data but the accuracy dropped significantly in test data. Which shows how that these models over fits the data.

VII. CONCLUSION

Satellite imagery provides a comprehensive and timely assessment of agricultural parameters such as crop health, land use, and vegetation indices, offering a data-rich perspective of agricultural landscapes

Satellite imagery offers a non-intrusive and scalable solution for collecting agricultural data, providing cost-effective and widespread coverage without the need for extensive fieldwork.

Satellites equipped with advanced sensors enable continuous monitoring of large agricultural areas, allowing for the tracking of changes over time, identification of trends, and prediction of crop yields or potential failures.

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